Changes of State

Activity 1: Match the key words to the statements. There are more key words than statements.

Solidification, Melting, Evaporation, Boiling, Heating, Condensation, Phase, Sublimation, Ice, Vapour, Water, Freezing, Solid, Gas, Cooling, Atoms, Liquid

1. Gas changes to liquid _____
2. Liquid changes to gas _____
3. Solid changes to liquid _____
4. Liquid changes to solid _____
5. Solid changes to gas _____
6. Speeds particles up _____
7. Slows particles down _____
8. Particles move freely in this state _____
9. Particles can only vibrate in this state _____
10. Particles can roll over each other in this state _____
11. Another word for state of matter _____
12. Particles that make up matter _____

Activity 2: Fill in the missing words in the text below. Words can be used more than once.

Heating, Solidified, Fixed, Particles, Position, Liquid, Original, Gas, Solid

All matter is made up of 1. _____. In a solid the 2. _____ are all stacked up in neat rows. Forces between them keep them 3. _____ in position as if they were all tied together with little springs. The atoms are always moving. This movement increases when the solid is heated. As the 4. _____ continues, the motion becomes so violent that all the little springs or bonds holding them together break. The atoms no longer stick together in neat stacks but wander around, never staying in the same 5. _____. The solid has melted and has become a 6. _____.

After more 7. _____, the atoms are moving so violently that they no longer stay together at all and fly off in all directions. The 8. _____ has boiled and become a gas.

We can turn the 9. _____ into a liquid by slowing down the atoms or molecules. This can be done by cooling the gas until it condenses into a 10. _____. In order to turn the liquid into a 11. _____ we need to slow the atoms or molecules down even further by more cooling. Once cooling has

slowed the 12. _____ enough for them to take up their 13. _____ positions the solid reforms. We then say the liquid has frozen or 14. _____.

Matter changing from one form into another is said to undergo a change of state. The changes of state can be classified into two groups; those that occur after 15. _____ and those that occur after cooling.

Activity 3: Unjumble the following sentences.

1. Solids constant shape a have.
2. Compressed gases can easily be.
3. Poured gases can be liquids and.
4. Fluids, gases and liquids called together are.
5. Matter all heated when is the expands it.
6. Shape the container liquids of take their.

Activity 4: Use the following words in a complete sentence so that the sentence explains what each word means.

1. Compress
2. Expand
3. State of matter

Activity 5: Answer the following questions using the particle theory of matter. Use diagrams to help explain your answers.

1. Solids cannot be compressed.

2. Liquids take the shape of their container.

3. Liquids cannot be compressed.

4. Solids cannot take the shape of their container.

5. Gases can be compressed.

6. Smells from cooking can be smelt throughout the house.